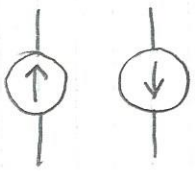
W2AEW

BACK TO BASICS:

TRANSISTOR CURRENT SOURCES



- FIXED CURRENT
- POSITIVE OR NEGATIVE
- HIGH OUTPUT IMPEDANCE

VS.

- FIXED VOLTAGE
- POSITIVE OR NEGATIVE
- LOW OUTPUT IMPEDANCE

SOME APPLICATIONS:

END FUNCTIONS

- LED DRIVER
- BATTERY CHARGER
- SENSOR DRIVER

TEST & MEASUREMENT

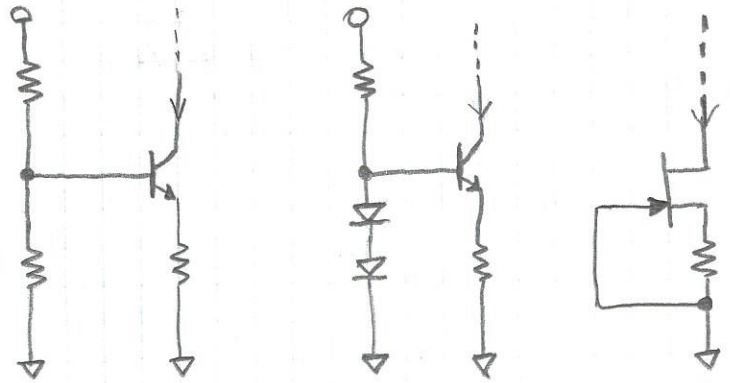
- RESISTANCE MEAS.
- DIODE TEST
- BRIDGE DRIVER

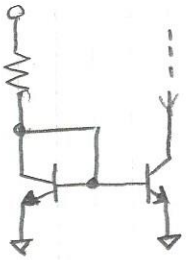
BUILDING BLOCKS

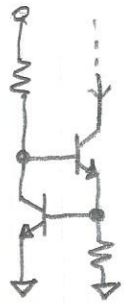
- BIAS CIRCUITS
- ACTIVE LOAD
- RAMP GENERATOR

MANY CONFIGURATIONS:

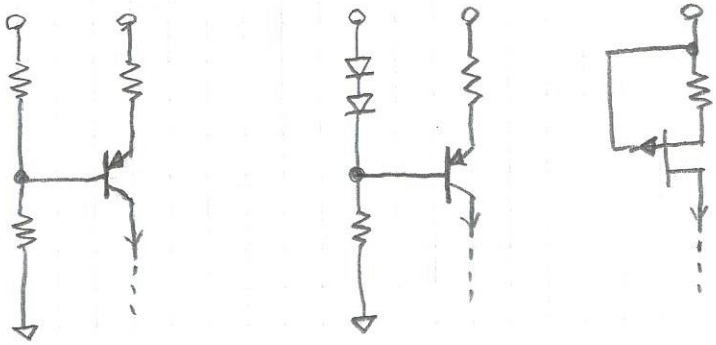
"SINK"







"SOURCE"



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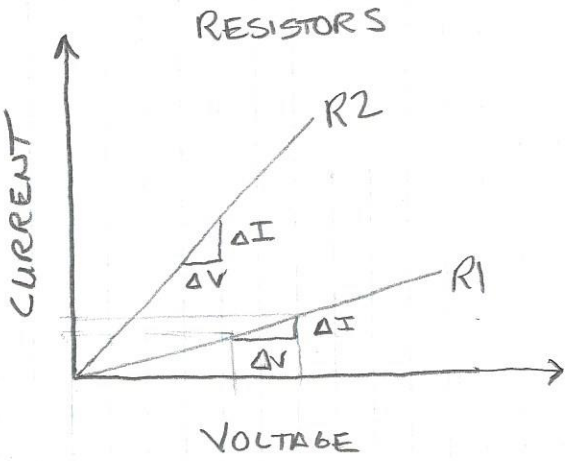
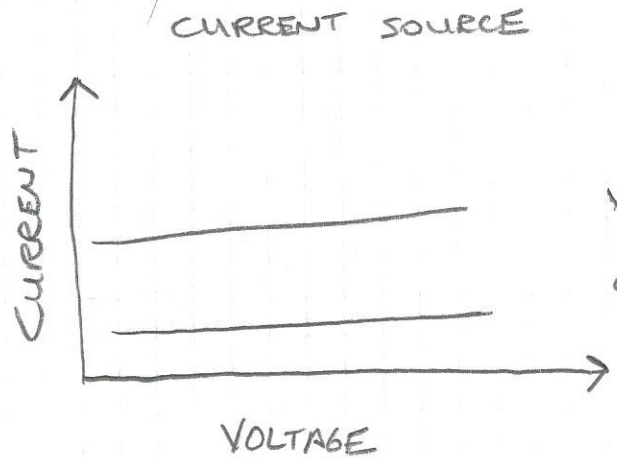
Output Impedance



OBEYS OHMS LAW

$$R = \frac{V}{I}$$



VERY LITTLE CHANGE IN CURRENT VS. VOLTAGE

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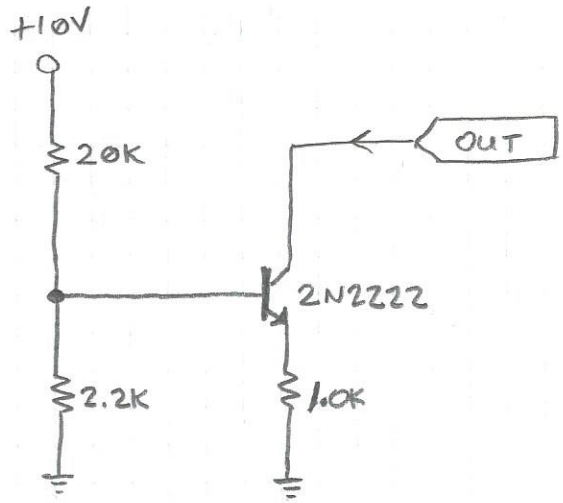
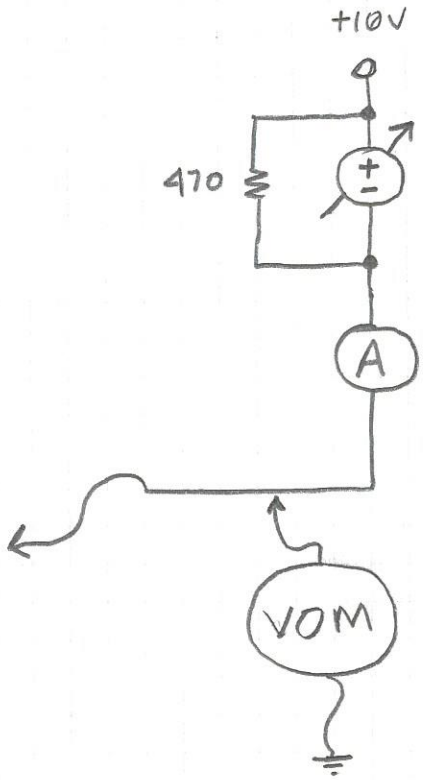
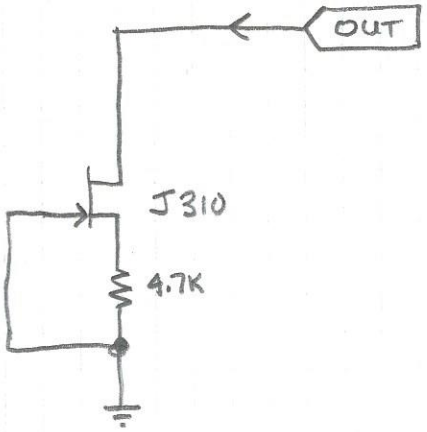
TRANSISTOR CURRENT SOURCE TEST CIRCUITS

10V - 2V

419μA - 377μA

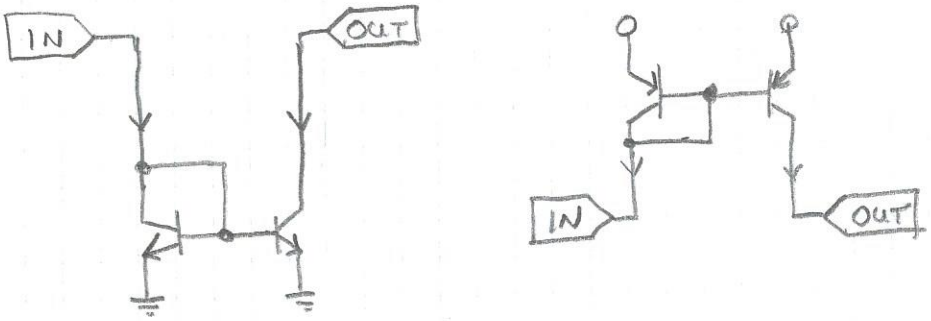
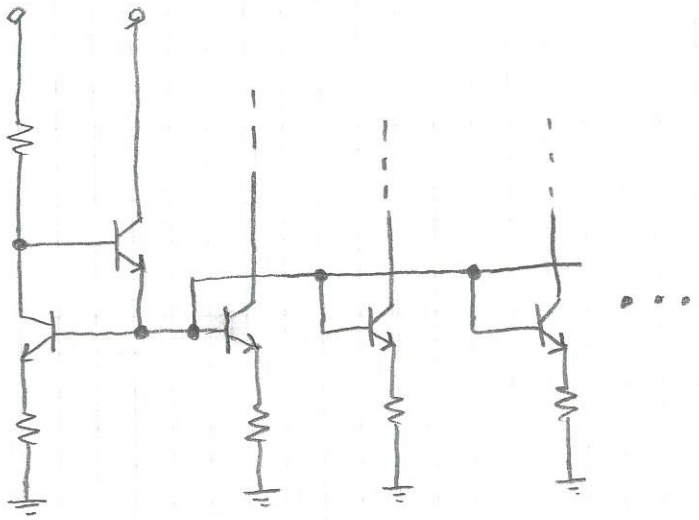






TURNING IT AROUND WITH A...

CURRENT MIRROR



STRING OF CURRENT SOURCES...